CHAPTER 1: MULTI-REGION CLUSTER ARCHITECTURE AND NETWORK TOPOLOGY

The cloud infrastructure is organized into a multi-region active-active topology spanning US-East, US-West, and
Each region connects via dedicated VPC peering with non-overlapping CIDR blocks:
- Primary Region (US-East): 10.120.0.0/16
- Secondary Region (US-West): 10.121.0.0/16
- Disaster Recovery Region (EU-Central): 10.122.0.0/16

All compute nodes are distributed across a minimum of 3 availability zones per region to maintain 99.995% avail
Redundant edge ingress gateways perform BGP anycast routing and terminate public TLS connections before tr
The maximum allowable network latency between availability zones within any single region is strictly capped at

CHAPTER 2: MUTUAL TLS SECURITY PROTOCOLS AND CERTIFICATE ROTATION

All internal microservice-to-microservice communication requires mutual TLS (mTLS) with strict cryptographic va
The only permitted cipher suite for internal mesh communications is ECDHE-ECDSA-AES256-GCM-SHA384 wit

Certificate Lifecycle Management:
- Standard certificate rotation window: 90 days.
- Automated certificate renewal trigger: 30 days prior to expiration date.
- Emergency certificate revocation: Executed via CRL distribution points within 60 seconds of compromise detec
- Identity Attestation: Every workload must prove identity using SPIFFE IDs formatted as spiffe://internal.mesh/ns

Any communication attempt lacking a valid SPIFFE credential or presenting an expired x509 certificate is rejecte

CHAPTER 3: DISASTER RECOVERY OBJECTIVES AND STORAGE BACKUP RTO/RPO

Business continuity and disaster recovery procedures define strict recovery metrics for tier-1 transactional syster
- Recovery Time Objective (RTO): Strictly under 15 minutes for complete regional failover.
- Recovery Point Objective (RPO): Strictly under 60 seconds of transaction data loss.

Storage Replication Architecture:
Data persistence layers utilize cross-region asynchronous snapshot replication streaming to immutable cold stora
Snapshots are captured automatically every 30 seconds and mirrored to EU-Central.

Automatic Failover Trigger Thresholds:
Failover is initiated if the primary cluster fails 5 consecutive health probes spanning a 120-second evaluation win
or if regional packet loss exceeds 45% for longer than 90 continuous seconds.
Manual override requires authorization from two designated engineering leads using dual-custody MFA tokens.